Math for Machine Learning

YouTube roadmap: from basics to advanced

Suggested order: Linear Algebra, Calculus, Statistics, then ML applications. Spend the most time on linear algebra and calculus, since they are used the most in practice.

1. Linear Algebra (vectors, matrices, eigenvalues, PCA)

3Blue1Brown: Essence of Linear Algebra [found in search]
16 videos. Best for intuition. Start here.
Open on YouTube

MIT 18.06 (Gilbert Strang) [found in search]
Full university course for depth.
Open on YouTube

Jon Krohn: Linear Algebra for ML
Complete course built for ML (search link).
Open on YouTube

2. Calculus (derivatives, gradients, backpropagation)

3Blue1Brown: Essence of Calculus
Intuition first.
Open on YouTube

Professor Leonard: Calculus 1 [found in search]
Practice and problem solving.
Open on YouTube

Multivariable Calculus [found in search]
Partial derivatives and gradients, needed for gradient descent.
Open on YouTube

3. Probability and Statistics

StatQuest: Statistics Fundamentals
Distributions, variance, confidence intervals, p-values, maximum likelihood.
Open on YouTube

4. Applying the Math to ML

StatQuest: Machine Learning
Regression, PCA, gradient descent, neural networks.
Open on YouTube

3Blue1Brown: Neural Networks
How gradients and backpropagation work.
Open on YouTube

Andrej Karpathy: Neural Networks, Zero to Hero
Hands-on deep learning from scratch (search link).
Open on YouTube

Items marked [found in search] were confirmed in web results. Other links are from general knowledge, so please check that they open correctly.